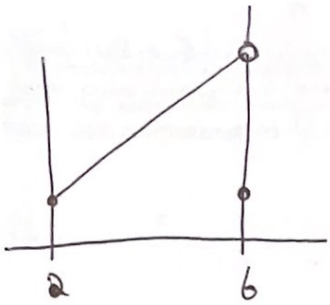
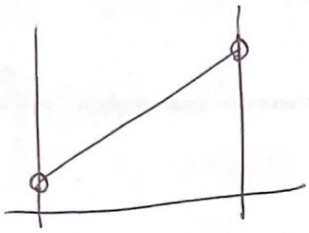
Exercise 1

a)



$$f(x) = \begin{cases} x & \text{if } x \neq b \\ a & \text{if } x = b \end{cases}$$

b)



$$f(x) = x$$

c)

$$f(x) = (x-a)\sin\left(\frac{1}{x-b}\right)$$

Exercise 2

a) By def of limit $\exists\ M_1$ s.t. $f(x) < f(0)\ \forall\ x > M_1$
and $\exists\ M_2$ s.t. $f(x) < f(0)\ \forall\ x < M_2$.
Pick $M = \max\{M_1, -M_2\}$.

b) We apply Weierstrass thm on f restricted to $[-M, M]$.
Hence $\exists\ x_0 \in [-M, M]$ s.t. $f(x_0) \geq f(x)\ \forall\ x \in [-M, M]$.
From b) we know that $f(x_0) \geq f(0) \geq f(x)\ \forall\ x \notin [-M, M]$.
In conclusion $f(x_0) \geq f(x)\ \forall\ x \in \mathbb{R} \Rightarrow f$ has a maximum.

c) f cannot have a minimum.
$\inf f = 0$ since $f(x) > 0\ \forall\ x$ and $\lim_{x \to +\infty} f(x) = 0$
But the inf can't be achived since $f(x) > 0\ \forall\ x \in \mathbb{R}$.

Exercise 3

a) $\left|e^x - 1 - x\right| = \left|\sum_{n=2}^{\infty} \frac{x^n}{n!}\right| \leq \sum_{n=2}^{\infty} \frac{|x|^n}{n!} \leq \sum_{n=2}^{\infty} \frac{|x|^n}{2^{n-1}}$

$\leq \sum_{n=2}^{\infty} \frac{1}{2^{n-1}} = 1$ if $|x| < 1$. (Can be deduced directly by reminder thm).

Then $\lim_{x \to 0} \left|\frac{e^x - 1}{x} - 1\right| = \lim_{x \to 0} \left|\frac{e^x - 1 - x}{x}\right|$

We look at $\left|\frac{e^x - 1 - x}{x}\right| \leq \frac{|x^2|}{|x|} \leq |x|$ for $|x| < 1$

$\lim_{x \to 0} |x| = 0$

And by squeeze thm $\lim_{x \to 0} \left|\frac{e^x - 1}{x} - 1\right| = 0$ as well

b) Let us define $f(x) = \begin{cases} \frac{x}{e^x - 1} & x \neq 0 \\ 1 & x = 0 \end{cases}$ and $g(x) = \log(1+x)$

$f: \mathbb{R} \to \mathbb{R}$ $\qquad g: (-1, \infty) \to \mathbb{R}$

We know from class that g is continuous at 0 and $g(0) = 0$, and from a) that f is continuous at $0 = g(0)$. Then by continuity of the composition we get

$1 = \lim_{x \to 0} f \circ g(0) = \lim_{x \to 0} f(g(x))$

$= \lim_{x \to 0} \frac{g(x)}{e^{g(x)} - 1} = \lim_{x \to 0} \frac{\log(1+x)}{x + 1 - 1} = \lim_{x \to 0} \frac{\log(1+x)}{x}$ as desired.

Exercise 4

a) $z_0 = 1+i \qquad |z_0| = \sqrt{1+1} = \sqrt{2} \qquad z_0^0 = 1 \qquad |z_0^0| = 1$

$z_0^2 = (1+i)(1+i) \qquad |z_0^2| = \sqrt{2^2} = 2$

$= 1 + 2i - 1 = 2i$

$z_0^3 = z_0^2 z_0^1 = 2i(1+i) \qquad |z_0^3| = \sqrt{2^2+2^2} = 2\sqrt{2}$

$= 2i - 2$

$z_0^4 = z_0^3 z_0^1 = (2i-2)(1+i) \qquad |z_0^4| = 4$

$= \cancel{2i} - 2 - 2 \cancel{- 2i} = -4$

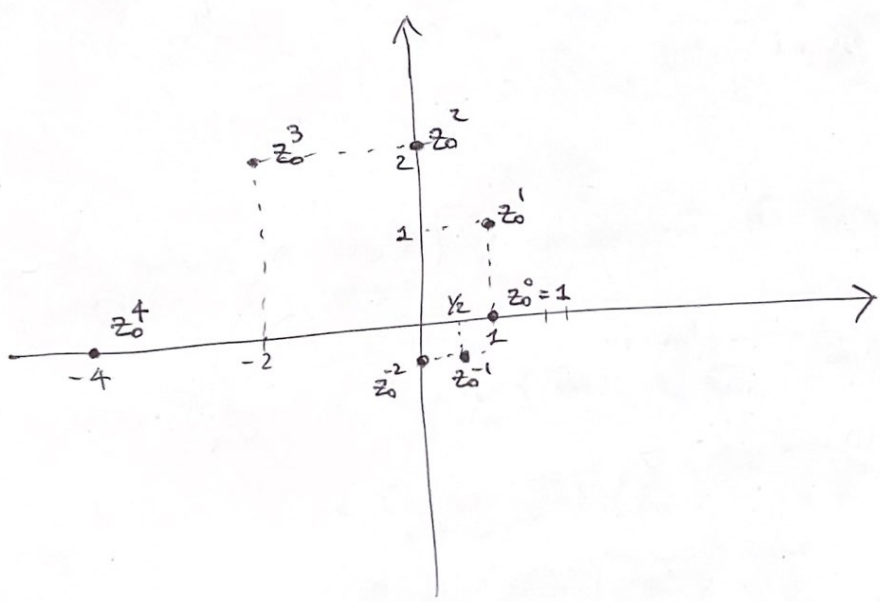
$\cancel{z_0^5 = z_0^4 z_0^1 = -4(i+1)} \qquad \cancel{|z_4| = \sqrt{16+16} = 4\sqrt{2}}$

$\cancel{= -4 - 4i}$

b) $z_0^{-1} = \frac{1}{i+1} = \frac{1}{i+1}\frac{1-i}{1-i} = \frac{1-i}{1^2-i^2} = \qquad |z_0^{-1}| = \sqrt{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\sqrt{2}}$

$= \frac{1}{2} - \frac{1}{2}i$

$z_0^{-2} = (z_0^{-1})^2 = \frac{1}{4} - \frac{1}{4} - \frac{1}{2}i = -\frac{1}{2}i \qquad |z_0^{-2}| = \frac{1}{2}$



Exercise 5

We first recall that $\lim_{n\to+\infty} \sqrt[n]{n} = 1$

(Proof (if needed): $\sqrt[n]{n} < 1+\varepsilon \iff n < (1+\varepsilon)^n$ and $\exists\, n_0 > 0$ s.t. ~~and~~ this is true for $n > n_0$ because exponential beats polynomial if the base is bigger than 1

$\sqrt[n]{n} > 1-\varepsilon \iff n > (1-\varepsilon)^n$ same thing but n wins when the base is smaller than 1).)

a) For the first series

$$R = \limsup_{n\to+\infty} \sqrt[n]{\frac{1}{a_n}} = \lim_{n\to+\infty} \sqrt[n]{\frac{1}{n^2 2^n}}$$

$$= \limsup_{n\to+\infty} \frac{1}{2}\left(\sqrt[n]{n}\right)^{-2} = \frac{1}{2}\left(\lim_{n\to+\infty} \sqrt[n]{n}\right)^{-2}$$

$$= \frac{1}{2}\cdot(1)^{-2} = \frac{1}{2}$$

For the second one we apply the ratio criterion:

$$\lim_{n\to+\infty} \frac{|z|^{n+1}}{(n+1)^{n+1}} \cdot \frac{(n+1)!}{|z|^n\, n!} \cdot \frac{(n+1)^n}{} = \lim_{n\to+\infty} |z| \cdot \frac{(n+1)^n}{(n+1)^n}$$

$$= \lim_{n\to+\infty} |z| \cdot \left[\left(1+\frac{1}{n}\right)^n\right]^{-1}$$

$$= \frac{|z|}{e}$$

Then the series is absolutely convergent if $|z| < e$, and divergent if $|z| > e$.

$\Rightarrow R = e$

b) $$R' = \limsup_{n\to+\infty} \sqrt[n]{\frac{1}{n\, a_{n-1}}} = \limsup_{n\to+\infty} \left(\sqrt[n]{n}\right)^{-1} \sqrt[n-1]{\frac{1}{a_{n-1}}} \cdot \left(\frac{1}{a_{n-1}}\right)^{+\left(\frac{1}{n}-\frac{1}{n-1}\right)}$$

$$= \limsup_{n\to+\infty} \left(\sqrt[n]{n}\right)^{-1} \left(\frac{1}{a_{n-1}}\right)^{\frac{1}{n(n-1)}} \sqrt[n-1]{\frac{1}{a_{n-1}}}$$

$$R := \limsup \sqrt[n]{\frac{1}{a_n}}$$

$$= \limsup \left(\sqrt[n]{n}\right)^{-1} \left(\sqrt[n-1]{\frac{1}{a_{n-1}}}\right)^{\frac{1}{n}} \sqrt[n-1]{\frac{1}{a_{n-1}}}$$

$$= \lim \left(\sqrt[n]{n}\right)^{-1} \limsup \left(\sqrt[n-1]{\frac{1}{a_{n-1}}}\right)^{\frac{1}{n}} \cdot \sqrt[n-1]{\frac{1}{a_{n-1}}}$$

$$= 1 \cdot 1 \cdot R = R$$